ELLIPTIC

SANCTIONS COMPLIANCE IN CRYPTOCURRENCIES:

Five keys to preparing your business with blockchain analytics

Executive Summary

Economic and financial sanctions are set to have a major impact on the cryptocurrency space.

In November 2018, the United States Treasury identified two specific bitcoin addresses belonging to sanctioned Iranian money launderers. The first listing of its kind, it put the cryptocurrency industry on notice that sanctions bodies are watching the space closely and are prepared to act to limit the ability of cryptocurrencies to facilitate global and national security threats.

With Russia, Iran, North Korea and Venezuela increasingly looking to cryptocurrencies to engage in illicit activity and evade international financial restrictions, additional sanctions actions impacting the cryptocurrency space are inevitable.

Cryptocurrency businesses must prepare for a tightening sanctions compliance environment.

Those that fail to take appropriate steps could now find themselves in regulators' crosshairs, risking large fines or penalties. **Elliptic research shows that approximately $2.4 million in cryptocurrency funds from the OFAC-sanctioned Iranian actors passed through major mainstream exchanges and wallet services, many in Europe and the US.**

Preparedness is key, and compliance officers must take a proactive approach. If the cryptocurrency industry is to flourish, businesses cannot ignore sanctions requirements, and must avoid interaction with the actors who are the target of sanctions actions.

At Elliptic, we empower financial institutions to do just that, by providing effective and easy-to-use solutions that make use of blockchain insights to identify and investigate risk in cryptocurrency transactions.

In this report, we explore how compliance officers can protect their businesses against exposure to cryptocurrency-related sanctions risks. In particular we make three key recommendations on the use of blockchain analysis insights:

1 It is not enough for businesses to simply check whether they have transacted directly with cryptocurrency addresses on sanctions lists. Funds should be traced through the blockchain to ensure they do not ultimately interact with these addresses, even indirectly.

2 Businesses must look beyond the specific addresses on sanctions lists - other addresses might exist that are in the same wallets and which are controlled by the same parties. Transactions should be checked for links with these additional addresses.

3 Businesses should also screen for any transactions, direct or indirect, with parties that are known to be located in sanctioned jurisdictions, even where the specific cryptocurrency addresses of those parties may not be explicitly called out by regulators.

Ensuring a successful response to sanctions actions will be an ongoing journey for the cryptocurrency industry in the years ahead. With this report, we've set out a blueprint for enabling cryptocurrency exchanges and other businesses in the space to manage that process successfully.

Introduction

On November 28, 2018, the US Department of the Treasury's Office of Foreign Assets Control (OFAC) undertook a milestone action when, for the first time, it added two bitcoin addresses to its list of Specially Designated Nationals (SDNs).

The two addresses were controlled by Ali Khorashadizadeh and Mohammad Ghorbaniyan, Iranian-based cryptocurrency brokers who moved funds for the perpetrators of the SamSam ransomware campaign, and who engaged in other cryptocurrency transactions totalling more than $17 million using the two OFAC-listed addresses alone.

> "Treasury Designates Iran-Based Financial Facilitators of Malicious Cyber Activity and for the First Time Identifies Associated Digital Currency Addresses"
>
> **- US DEPARTMENT OF THE TREASURY PRESS RELEASE, NOVEMBER 28, 2018**

OFAC SDN LIST

KHORASHADIZADEH, Ali (a.k.a. "Iranvisacart"; a.k.a. "Mastercartaria"), Iran; DOB 21 Sep 1979; POB Tehran, Iran; nationality Iran; Email Address iranvisacart@yahoo.com; alt. Email Address mastercartaria@yahoo.com; alt. Email Address alikhorashadi@yahoo.com; alt. Email Address toppglasses@gmail.com; alt. Email Address iranian_boy5@yahoo.com; Additional Sanctions Information - Subject to Secondary Sanctions; Gender Male; Digital Currency Address - XBT 149w62rY42aZBox8fGcmqNsXUzSStKeq8C; Passport T14553558 (Iran) issued 28 Oct 2008 expires 29 Oct 2013 (individual) [CYBER2].

GHORBANIAN, Mohammad (a.k.a. GHORBANIYAN, Mohammad; a.k.a. "EnExchanger"; a.k.a. "Ensaniyat"; a.k.a. "Ensaniyat_Exchanger"), Iran; DOB 09 Mar 1987; POB Tehran, Iran; nationality Iran; Website www.enexchanger.com; Email Address EnExchanger@gmail.com; alt. Email Address Ensaniyat1365@gmail.com; Additional Sanctions Information - Subject to Secondary Sanctions; Gender Male; Digital Currency Address - XBT 1AjZPMsnmpdK2Rv9KQNfMurTXinscVro9V; Identification Number 008-046347-9 (Iran); Birth Certificate Number 32270 (Iran) (individual) [CYBER2].

The listing signalled OFAC's formal entry into the cryptocurrency space, and sent a clear warning: the cryptocurrency industry must be fully prepared to navigate the complex challenges of sanctions compliance, just as the banking, insurance, shipping, and other industries have done for years.

But the recent OFAC action carries a further set of lessons for compliance officers in the cryptocurrency space.

Before their OFAC listing, Khorashadizadeh and Ghorbaniyan operated from Iran and engaged in bitcoin transactions with nearly three dozen cryptocurrency exchanges around the world. These include exchanges in the US, Europe and Asia that deploy Know Your Customer (KYC) and other anti-money laundering (AML) controls. Our analysis shows that even legitimate exchange and wallet platforms in these regions handled as much as $2.4 million in funds from these Iran-based actors. Many of these exchanges were therefore exposed to Iran-related sanctions risks without being aware of it - a testament to the importance of ensuring a comprehensive and proactive approach to sanctions risk management and detection.

While the November 2018 bitcoin address listings may have been OFAC's first major action in the crypto-sphere, they surely will not be the last.

The emergence of rogue state actors and their affiliates in the crypto space over the past year is likely to keep cryptocurrencies squarely in regulators' sights. For example:

- Russia has used cryptocurrencies to engage in espionage[1].
- North Korea has engaged in cryptocurrency-enabled cybercrime to raise funds[2].
- Venezuela has launched its own cryptocurrency to facilitate sanctions evasion[3].
- Iran has reportedly used cryptocurrencies and has declared its intention, like Venezuela, to use them for facilitating transactions outside the traditional financial sector.

All signs therefore point to a tightening sanctions regulatory posture that will have a major impact on the cryptocurrency compliance space.

> "We are publishing digital currency addresses to identify illicit actors operating in the digital currency space. Treasury will aggressively pursue Iran and other rogue regimes attempting to exploit digital currencies and weaknesses in cyber and AML/CFT safeguards to further their nefarious objectives."
>
> **- US TREASURY UNDER SECRETARY FOR TERRORISM AND FINANCIAL INTELLIGENCE SIGAL MANDELKER, NOVEMBER 2018**

And just as other sectors have seen fines and penalties imposed for sanctions violations, cryptocurrency exchanges should not expect to be treated with kid gloves.

[1] https://www.elliptic.co/our-thinking/doj-indictment-russian-hackers-blockchain-analysis
[2] https://www.elliptic.co/our-thinking/following-money-from-bithumb-hack
[3] https://www.elliptic.co/our-thinking/venezuelan-petro-sanctions-scams-smokescreens

Between January 2003 and March 2019, OFAC levied civil penalties for sanctions violations totalling more than $4.3 billion.[4] This figure includes an enormous penalty of $963 million imposed on BNP Paribas in 2014 for violating US restrictions on dealings with sanctioned countries. Companies in the fintech space, such as PayPal, have faced regulatory action by US sanctions authorities as well - and there's little reason to think that cryptocurrency exchanges that engage in violations won't run a similar risk.

Even cryptocurrency businesses outside the US need to be alert to the risk, as they can face secondary sanctions for facilitating business with US-listed entities, or penalties for causing violations of US sanctions.

> "PayPal to Pay $7.7 Million to U.S. Over Alleged Sanctions Violations"
>
> **- WALL STREET JOURNAL MARCH 25, 2015**

Amid a rapidly evolving threat landscape, and with regulators determined not to allow cryptocurrencies to provide a safe haven for rogue actors, compliance officers at cryptocurrency exchanges must not be complacent. According to a UN Security Council panel of experts, North Korea may have acquired as much as $500 million in cryptocurrencies through exchange hacks and other methods.[5] As sanctioned actors increasingly interact with the crypto space, compliance officers need to be alert to the likelihood of increased exposure to these parties.

In this report, we set out five key steps your business can take to navigate the emerging challenge of sanctions compliance in cryptocurrencies with success. Those are:

1. **Deploying the right monitoring tools.** Have you deployed transaction monitoring solutions that enable effective risk-based monitoring and rely on best-in-class data that ensure you can detect connections to sanctioned parties?
2. **Managing your country risk exposure.** Are you able to identify more subtle signs of sanctions risks, such as potential exposure to entities located in or near sanctioned jurisdictions?
3. **Knowing the red flags.** In addition to geographical risk indicators, are your staff aware of red flags and suspicious indicators that may be indicative of high risk activity that may carry sanctions risks?
4. **Defining your investigative strategy.** Where risks have been identified, are you equipped to investigate potential sanctions breaches and report them to the appropriate authorities?
5. **Embedding a comprehensive risk management framework.** Have you conducted a sanctions risk assessment to measure your overall level of risk exposure, and have you designed the processes and procedures necessary to mitigate that risk?

Below are five keys to achieve these goals and make your company's cryptocurrency sanctions compliance journey as smooth as possible.

[4] https://www.treasury.gov/resource-center/sanctions/CivPen/Pages/2018.aspx
[5] https://www.undocs.org/en/S/2019/171



5 Keys to a Successful Cryptocurrency Sanctions Compliance Response

1. Deploying Effective Blockchain Monitoring Tools

Having the right technical solutions in place ensures you avoid exposure to sanctioned entities and individuals that use cryptocurrencies.

Correctly utilising the solutions we have developed at Elliptic, which rely on best-in-class data quality, enables you to engage in risk-based monitoring and to detect potential connections to sanctioned parties with confidence.

The November 2018 OFAC action is notable not only because it was the first time cryptocurrency addresses were called-out for sanctions purposes. By listing specific addresses belonging to known facilitators of illicit cryptocurrency activity, the U.S. Treasury provided Elliptic with the clues we needed to be able to understand in detail how these actors operate.

Elliptic's response to the OFAC action was swift: we immediately updated our systems to clearly label the two OFAC-listed addresses. What's more, we detected two additional bitcoin addresses in the same wallet as the OFAC-listed addresses, but which OFAC hadn't explicitly mentioned in its action. This is significant since all of these addresses can be associated with the individuals on the SDN list. If you are unaware of these additional addresses you run the risk of unknowingly transacting with these individuals.

Including these four addresses in our tool has enabled compliance officers who use our AML software to identify potential links to the sanctioned persons and identify historical activity of concern.

It also enables us to learn a tremendous amount about how Khorashadizadeh and Ghorbaniyan were operating.

By examining bitcoin blockchain data, we can see that they were prolific bitcoin users who engaged in thousands of transactions over the course of several years to move funds, before being added to the OFAC SDN list. Methods used included:

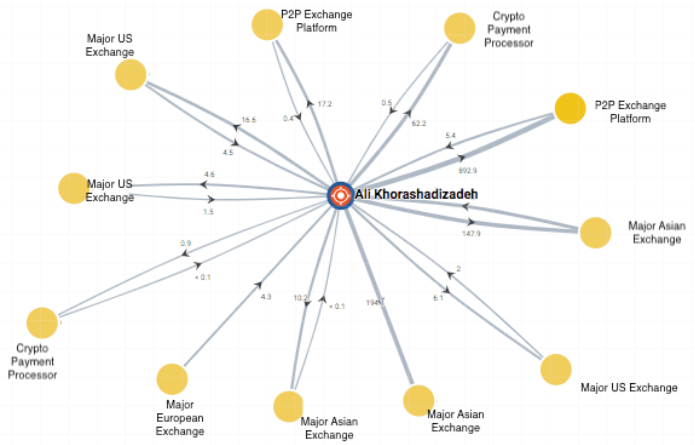
- employing the now-defunct BTC-e, which was a favoured exchange for global criminals, to swap cryptocurrencies;
- using peer-to-peer trading platforms such as LocalBitcoins and Paxful to facilitate business;
- using dozens of compliant exchanges in the US, Europe, and Asia;
- relying on cryptocurrency payment processing services in the US and Europe to make direct purchases for items using bitcoin;
- the use of cryptocurrency debit card services;
- moving funds via gambling sites that accept cryptocurrencies; and
- using at least one decentralised exchange (DEX) platform.

Not only does this activity demonstrate that OFAC hit the mark by targeting adept and prolific users of cryptocurrencies, but it indicates that all types of cryptocurrency platforms - even those that strive to be compliant - must be alert to the risk of exposure to sanctioned parties.

As the image below shows, prior to his designation by OFAC, Khorashadizadeh transacted with P2P exchange platforms, centralized exchanges, and crypto payment processors, many of them outside Iran. Listing his bitcoin address will ensure that many of those platforms do not interact with that address again.

None of this is to say that sanctions actions targeting these activities are fool-proof. Reporting suggests that Ghorbaniyan has used Perfect Money, a centralized online value transfer system, to skirt sanctions, and he also claims to have created a new bitcoin address that has not been listed publicly.[6]

Regardless, having the ability to monitor potential interactions with the two OFAC-listed entities is a critical step in any cryptocurrency business's sanctions compliance journey.



This diagram demonstrates flows of bitcoin between Ali Khorashadizadeh and major crypto platforms across the world between 2013 and 2018. The figures represent the number of bitcoin that flowed to and from these entities from Khorashadizadeh.

Monitoring cryptocurrency transactions for potential sanctions exposure comes with certain specific challenges.

- Because cryptocurrency transactions can be highly transparent, it is possible to observe indirect connections to sanctioned parties. Because of this visibility, and the peer-to-peer nature of crypto payments, it is not enough to take the traditional approach and simply screen your counterparties' names or cryptocurrency addresses against sanctions lists. A sanctioned entity can simply pass crypto funds through several new wallets before sending them to a financial institution. Luckily, this activity is visible on the blockchain and Elliptic's software can be used to quickly determine whether a transaction ultimately originates from a sanctioned entity.

- A cryptocurrency exchange may, for instance, have a customer whose transactional activity is linked to an OFAC-sanctioned address through many "hops" involving an OFAC-sanctioned address through many "hops", i.e. funds may have passed through other wallets. Identifying these indirect connections is vital to ensuring sanctions compliance, but determining whether a violation occurred is not always straightforward. It is not enough to look for direct transactions between a customer and a sanctioned person's crypto address. Rather a compliance team must weigh a number of factors when making such a decision, including:

  1. the number of hops between the customer's address and the sanctioned party's address, and the velocity of flows between addresses in the transaction chain;
  2. the proportion of a customer's funds that may have flown indirectly to or from a sanctioned party's address; and
  3. whether the customer's broader transactional activity may include direct or indirect connections to entities featuring other high risk indicators, such as transactions with exchanges known to be located in sanctioned jurisdictions.

- A sanctioned party is likely to have control of cryptocurrency addresses other than those that have been specifically identified by OFAC or other sanctions authorities. For example, an exchange customer may send funds to a cryptocurrency address that Elliptic's software identifies as part of a wallet that also contains an OFAC-listed address, but that has not itself been explicitly identified by OFAC. Cryptocurrency exchanges must be able to identify all addresses controlled by sanctioned parties, whether identified directly by OFAC or not, to ensure they do not inadvertently process prohibited payments.

Blockchain monitoring tools are a critical tool to ensure sanctions compliance when handling cryptocurrencies.

[6] See: https://brief.kharon.com/updates/iranian-cryptotrader-implicated-in-ransomware-scheme-turning-to-mysterious-payment-system/

Case Study: How Blockchain Monitoring Shed Light on North Korea's Hack of a Cryptocurrency Exchange

In June 2018, the South Korean exchange Bithumb was the target of a significant cryptocurrency hack.

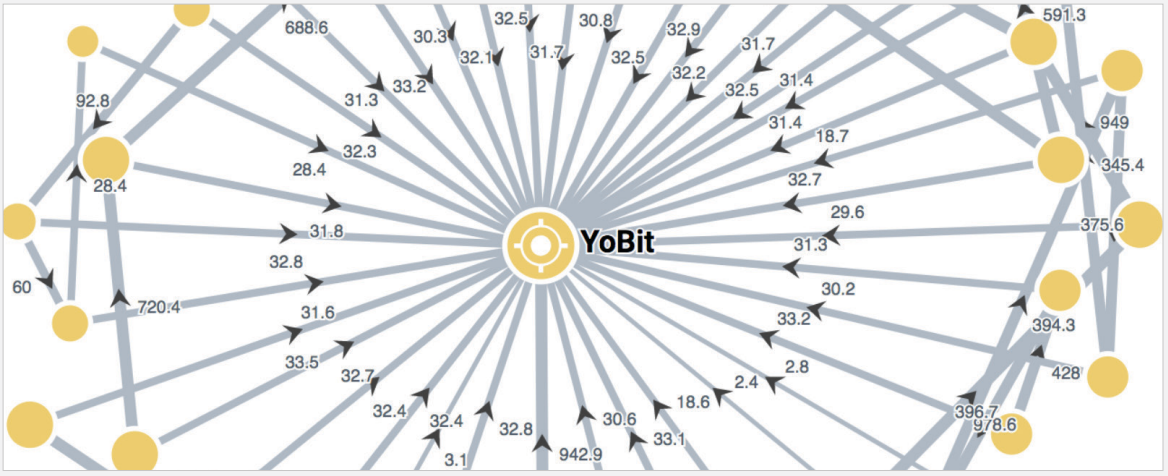
Cybercriminals managed to steal cryptocurrencies totalling $30 million from Bithumb. The attack has since been attributed by a cybersecurity intelligence firm to the North Korea-linked Lazarus Group of hackers, who were also responsible for the WannaCry ransomware attack in May 2017.[7]

At Elliptic, we were able to use our proprietary software to follow the flow of nearly $13 million worth of bitcoin that the hackers had taken from Bithumb.

Our analysis indicated that after stealing the funds from Bithumb, by making over 400 separate withdrawals to their own wallet, the hackers moved the funds to the Russia-based cryptocurrency exchange YoBit. which is presently unregulated and allows users to swap cryptocurrencies for fiat currencies as well as other digital services such as WebMoney and PerfectMoney. The stolen bitcoins were deposited at YoBit in 68 separate transactions, using a process known as 'chain-peeling'.

Chain-peeling involves repeatedly depositing unspent bitcoin into unused addresses - a technique that is designed to obscure the connection to the original user, with the hopes of obfuscating the transaction trail.

However, as the image below shows, our solutions enable us to track this activity, making it harder for sanctioned parties to hide. Our ability to track complex transactions can empower compliance officers to have visibility into activity that might otherwise go undetected.



Funds stolen from Bithumb flow into the YoBit exchange

"North Korean Hackers Make Millions from Cryptocurrency Exchanges"

- BITCOIN AUSTRALIA OCTOBER 22, 2018

[7] Chris Doman, "Malicious Documents From Lazarus Group Targeting South Korea," Alien Vault, June 22, 2018, https://www.alienvault.com/blogs/labs-research/malicious-documents-from-lazarus-group-targeting-south-korea

2. Managing Your Risk Exposure to Sanctioned Countries

Avoiding sanctions risks is about more than just monitoring for connections to specific SDNs or other known illicit actors.

A successful risk-mitigation strategy also involves detecting more subtle signs of risk, such as exposure to high risk countries, or to regions that might pose high risks of sanctions evasion activity.

For example, because US sanctions generally prohibit US financial institutions from dealing with Iranian financial institutions[8], cryptocurrency businesses need to be alert not only to interactions with Iran-based SDNs, but also to potential interactions with cryptocurrency exchanges and other services in Iran that have not been explicitly listed by OFAC.

> "Institutions should consider reviewing blockchain ledgers for activity that may originate or terminate in Iran."
>
> **- US FINANCIAL CRIMES ENFORCEMENT NETWORK, OCTOBER 2018**

Similarly, a recent US Executive Order prohibits US persons from having dealings involving any Venezuelan government-backed cryptocurrencies, a response to Venezuela's launch of the Petro cryptocurrency in December 2017.[9]



In April 2018, the Venezuelan government announced that it had approved 16 cryptocurrency exchanges domestically to handle the Petro.[10] Cryptocurrency exchanges outside Venezuela therefore need to be alert to potential connections to these exchanges, such as customers who may frequently utilize those exchanges, in order to mitigate their sanctions risk exposure.

> "U.S. bans transactions with Venezuela's digital currency"
>
> **- REUTERS MARCH 19, 2018**

[8] See Executive Order 13599 of February 5, 2012, "Blocking Property of the Government of Iran and Iranian Financial Institutions," https://www.treasury.gov/resource-center/sanctions/Programs/Documents/iran_eo_02062012.pdf
[9] See Executive Order 13827 of March 19, 2019, "Taking Additional Steps to Address the Situation in Venezuela," https://www.treasury.gov/resource-center/sanctions/Programs/Documents/13827.pdf
[10] Aziz Abdel-Qader, "16 Cryptocurrency Exchanges Get Approval to Launch in Venezuela, List Petro," Finance Magnates 30 April 2018, https://www.financemagnates.com/cryptocurrency/news/16-cryptocurrency-exchanges-get-approval-launch-venezuela-list-petro/

Case Study: Third Country Sanctions Evasion Risk

Sanctioned actors frequently target third countries as go-betweens to move funds and avoid scrutiny. Iranian sanctions evaders have frequently looked to countries such as Turkey, Lebanon, and the UAE to avoid US scrutiny. Both Iran and North Korea have utilized financial institutions in countries such as China, Malaysia, Singapore and elsewhere to elude both US and international restrictions.

Blockchain analysis of the two Iranian OFAC-listed addresses indicate that Khorashadizadeh and Ghorbaniyan engaged in transactions with entities in third countries that have been used in historical sanctions evasion activity. Their activity included dealings with:

- at least three exchanges based in Turkey;
- several exchanges with operations in Southeast Asia; and
- several exchanges based in China.

This activity suggests exchanges in these third countries need to be alert to the risks of sanctions-related activity. And exchanges located elsewhere in the world need to be alert to activity involving third country exchanges that could be high risk, where such activity appears in conjunction with other sanctions-related red flags.

3. Knowing the Red Flags

Because sanctioned individuals and entities go to great lengths to conceal their activity, it is essential that you also know what red flags to look out for and your compliance teams are trained in how to identify them.

Red flags of potential sanctions-related activity can involve both transactional behaviours, as well as a range of other qualitative indicators.

Normally, it is a combination of several red flags that should alert your compliance teams to sanctions, prompting them to take a closer look.

Below we outline some red flags that can be indicators of sanctions-related activity.

Cryptocurrency and Sanctions Risks - Key Red Flags

1. A customer attempts to log-on to an exchange using IP addresses, email addresses, phone numbers, or other identifying indicators registered in a sanctioned jurisdiction.
2. A customer is identified as being associated with advertisements for cryptocurrency brokerage activity on P2P trading sites available to users in sanctioned jurisdictions.
3. A customer engages in indirect transactions with exchanges in sanctioned jurisdictions to a degree that can't be logically explained.
4. A customer sends funds to a cryptocurrency address that forms part of a "cluster" of addresses (or wallet) associated with an OFAC-listed address, but that has not itself been identified by OFAC.
5. A customer frequently engages in transactions through or with entities in countries known to be associated with sanctions evasion activity, with no clear purpose or rationale for the activity in question.

4. Defining Your Investigative Strategy

If your compliance team identifies red flags that may suggest you have sanctions exposure, it is necessary to dig deeper.

You need to have in place an investigations strategy that allows you to look in depth at customer activity and exhaustively scrutinize it.

This is especially important in sanctions-related cases, where even indirect and seemingly remote connections between customers and sanctioned parties can carry severe regulatory consequences.

A well-designed investigative strategy includes:

1. ensuring that all relevant staff are trained in conducting cryptocurrency investigations;
2. having documented investigative procedures and recordkeeping policies in place;
3. leveraging network analysis and case management tools effectively;
4. having in place internal escalation processes for raising alerts where positive hits have been identified; and
5. clearly documenting investigation findings in final reports that can be shared with relevant regulatory bodies, law enforcement, or other relevant stakeholders.

Case Study: Russian Hackers and the US Election

In July 2018, the US Department of Justice unsealed an indictment against agents of Russia's Main Intelligence Directorate (GRU) who allegedly engaged in cyber attacks against the Democratic National Committee in an attempt to undermine the 2016 US presidential election process.

Earlier, in March 2018, OFAC sanctioned the individual GRU members who took part in the hack, and also put sanctions on related companies that they operated.

> "Treasury Sanctions Russian Cyber Actors for Interference with the 2016 U.S. Elections and Malicious Cyber-Attacks"
>
> **- TREASURY.GOV**

According to the DOJ's indictment, the GRU agents used cryptocurrencies, including bitcoin, to facilitate the hack and related activities. The indictment indicates that the Russian agents attempted to avoid contact with the formal financial system by using cryptocurrencies to purchase web hosting and other related services, and even mined cryptocurrencies for their own use.[11] The indictment describes a specific bitcoin transaction that occurred on 1 February 2016, when one operative instructed another to send .026043 bitcoins to a specific bitcoin address.

With just this small piece of information, we can use Elliptic's Forensic software to glean additional information. We can see, for example, that the funds used to facilitate this transaction originated with a cryptocurrency exchange based in Europe that allows the exchange of US Dollars, Euros, and Russian Ruble.[12]

We can also observe that the bitcoin addresses associated with the GRU hack were used to identify numerous other services via cryptocurrency payment processors and exchanges located in the US.

By identifying these connections, we're able to assist exchanges in investigating and understanding any exposure to high risk activity, enabling them to close accounts, fulfill reporting obligations, and develop controls to mitigate exposure to similar risks in the future.

[11] See: https://www.justice.gov/file/1080281/download
[12] Tom Robinson, "How the DOJ Indictment of Russian Hackers Is Supported by Blockchain Analysis," July 24 2018, Elliptic, https://www.elliptic.co/our-thinking/doj-indictment-russian-hackers-blockchain-analysis

5. Embedding a Comprehensive Risk Management Framework

The steps outlined above are essential, but they can only excel where they are supported by a comprehensive compliance framework for managing sanctions risks holistically.

A comprehensive sanctions compliance risk management framework includes:

1. conducting an enterprise-wide risk assessment to determine the extent of potential sanctions-risk exposure across customer, product, and market segments;
2. utilising effective sanctions list screening tools and ensuring those are effective in monitoring for hits against OFAC and other sanctions lists;
3. having training programs in place to ensure that key members of staff understand sanctions obligations, risks, and appropriate responses; and
4. developing policies and procedures that clearly define staff responsibilities and set out well-defined prohibited activities.

> "While there may not be a "one-size-fits-all" sanctions compliance program that can be universally adopted, especially since U.S. economic sanctions can apply to all types of industries and businesses, we do believe there are commonalities of a good program. This is particularly the case now when there is an increase in the use of sanctions across a broader spectrum of jurisdictions and programs."
>
> **- US TREASURY UNDER SECRETARY FOR TERRORISM AND FINANCIAL INTELLIGENCE SIGAL MANDELKER, DECEMBER 2018**

Conclusion

Sanctions compliance is by no means a simple task.

A rapidly evolving threat landscape and increasing scrutiny from regulators makes it all but certain that the sanctions-related challenges facing the cryptocurrency industry will only grow in complexity over time.

But if the cryptocurrency industry is to mature, compliance officers must face these challenges head-on and must navigate them successfully. Failure to do so can result in significant penalties and regulatory censure that cryptocurrency businesses can't afford to face.

By focusing on achieving the objectives outlined in this report, cryptocurrency compliance officers can ensure their sanctions compliance process is as smooth as possible.

About Us

Elliptic empowers financial institutions and cryptocurrency businesses across the world to confidently manage anti-money laundering risk and detect criminal activity on the blockchain. To date, clients have used Elliptic products to assess risk on several trillion dollars worth of crypto transactions, and uncover activity relating to money laundering, terrorist fundraising, fraud and other financial crime. Elliptic software is recognized as the global standard for regulatory compliance in the cryptocurrency industry. To learn more, visit www.elliptic.co.

About Our Authors

David Carlisle is Head of Community at Elliptic, where he leads outreach with regulators and other external stakeholders on crypto-specific compliance issues. David has over a decade of public and private sector experience working on sanctions compliance topics. From 2007 to 2010 he worked at OFAC as an analyst responsible for the design and roll-out of US sanctions programs targeting countries such as Myanmar, Iran, North Korea, and Sudan. Prior to joining Elliptic he also worked as a consultant advising major international financial institutions on sanctions compliance matters.

Dr. Tom Robinson is the Chief Scientist and Co-Founder of Elliptic. Tom has advised governments, tax authorities, and regulators on cryptocurrencies, and his forensic analysis of cryptocurrency transactions has been used to help secure the convictions of cybercriminals. He is also involved in the development of Elliptic's blockchain intelligence products, which are used by cryptocurrency businesses and financial institutions around the world to asses risk on cryptocurrency transactions. Tom is based in Elliptic's New York office.